WUZZUF: Power Query Transformation Steps Documentation

By

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This document details all transformation steps applied to the Wuzzuf job-postings dataset (`Wuzzuf-Jobs-Posting.csv`) in Power BI using Power Query Editor for the JobPulse Dashboard project. Each step includes a description, the action performed, and a placeholder for a screenshot. The transformations prepare the data for analysis in Power BI, creating a main `Wuzzuf-Jobs-Posting` table and a `Skills` table.

Load the Dataset

Import the raw `Wuzzuf-Jobs-Posting.csv` file into Power BI Power Query to begin transformations.

Steps:

- Opened Power BI Desktop.
- Clicked **Home** > **Get Data** > **Text/CSV**.
- Selected `Wuzzuf-Jobs-Posting.csv` from the local directory.
- Clicked **Transform Data** to open the Power Query Editor.
- Verified that all columns (e.g., `Job Posting ID`, `Job Title`, `Job Skills`, `Years of Experience`, `Number of Applicants`) loaded correctly.

Screenshot Placeholder:

Job Posting ID	Job Posting Date	Job Title	Job Title Full	Job Title Additional Info	Job Position Type	Job Position Level	Years of Experience	Job Location
2671714552	Monday, January 9, 2017	Software Engineer	Software Engineer	-	Full-time	Mid-Senior level	8	Ellicott City, MD
2681782167	Wednesday, January 25, 2017	Software Engineer	Software Engineer	-	Full-time	Associate	3	United States
2689411099	Thursday, April 6, 2017	Software Engineer	Software Engineer	-	Full-time	Mid-Senior level	6	San Francisco Bay Area
2684663290	Wednesday, May 17, 2017	Software Engineer	Software Engineer	-	Full-time	Entry level	1	Cypress, CA
2749916088	Sunday, October 15, 2017	Software Engineer	Software Engineer	-	Full-time	Mid-Senior level	5	New York City Metropolitan
2687897090	Monday, November 13, 2017	Software Engineer	Software Engineer	-	Full-time	Mid-Senior level	4	United States
2749762142	Tuesday, January 2, 2018	Software Engineer	Software Engineer	-	Full-time	Mid-Senior level	9	United States
2744662085	Sunday, January 21, 2018	Software Engineer	Software Engineer	-	Full-time	Associate	4	New York, United States
2712325477	Friday, February 2, 2018	Software Engineer	Software Engineer	-	Full-time	Mid-Senior level	7	Richardson, TX
2684137164	Monday, February 12, 2018	Software Engineer	Software Engineer	-	Full-time	Mid-Senior level	4	Raleigh-Durham-Chapel Hill
2744433838	Sunday, April 8, 2018	Software Engineer	Software Engineer	-	Full-time	Entry level	1	Wakefield, MA
2721725730	Wednesday, April 25, 2018	Software Engineer	Software Engineer	-	Full-time	Mid-Senior level	8	United States
2708483200	Tuesday, May 15, 2018	Software Engineer	Software Engineer	-	Full-time	Entry level	2	San Francisco, CA
2577809064	Tuesday, June 26, 2018	Software Engineer	Software Engineer	-	Full-time	Entry level	1	Atlanta, GA
2706514927	Tuesday, July 17, 2018	Software Engineer	Software Engineer	-	Full-time	Mid-Senior level	6	United States
2720258469	Thursday, August 9, 2018	Software Engineer	Software Engineer	-	Full-time	Entry level	1	Ashburn, VA
2677440672	Tuesday, August 14, 2018	Software Engineer	Software Engineer	-	Full-time	Mid-Senior level	5	United States
2678417907	Wednesday, October 3, 2018	Software Engineer	Software Engineer	-	Full-time	Mid-Senior level	4	United States
2710742817	Monday, October 8, 2018	Software Engineer	Software Engineer	-	Full-time	Entry level	2	Atlanta, GA
2713350321	Wednesday, November 14, 2018	Software Engineer	Software Engineer	-	Full-time	Mid-Senior level	7	Raleigh, NC
2688211330	Sunday, December 9, 2018	Software Engineer	Software Engineer	-	Full-time	Mid-Senior level	4	Florida, United States
2678477127	Monday, December 10, 2018	Software Engineer	Software Engineer	-	Full-time	Associate	4	Seattle, WA
2686673061	Friday, December 14, 2018	Software Engineer	Software Engineer	-	Full-time	Mid-Senior level	0	Bellevue, WA
2719784660	Saturday, December 15, 2018	Software Engineer	Software Engineer	-	Full-time	Entry level	0	Cambridge, MA
2597346734	Thursday, December 27, 2018	Software Engineer	Software Engineer	-	Full-time	Mid-Senior level	6	Jacksonville, FL
2713349492	Wednesday, February 20, 2019	Software Engineer	Software Engineer	-	Full-time	Mid-Senior level	6	Greater Boston
2646791041	Thursday, March 21, 2019	Software Engineer	Software Engineer	-	Full-time	Entry level	2	Rockville, MD

Remove Blank Rows

Remove any completely blank rows to ensure data quality and avoid errors in downstream calculations.

Steps:

- In Power Query Editor, right-clicked the table.
- Selected **Remove Rows** > **Remove Blank Rows**.
- Checked the row count to confirm rows were removed.

Handle Missing Values in Numerical Columns

Replace null values in numerical columns (`Years of Experience`, `Number of Applicants`) with 0 to ensure accurate calculations.

Steps:

- Selected `Years of Experience` column.
- Right-clicked > **Replace Values** > replace `null` with `0`.
- Repeated for `Number of Applicants`.
- Verified no nulls remained using filters.

Filter Out Nulls in Critical Columns

Remove rows where critical columns (`Job Posting Date`, `Job Title`, `Job Position Level`) are null to maintain data integrity.

Steps:

- Filtered `Job Posting Date`, `Job Title`, and `Job Position Level` columns.
- Unchecked `(null)` values from the filter options.

Clean Job Skills Column (Remove Brackets and Quotes)

Description: Clean the `Job Skills` column by removing square brackets (`[]`) and quotes (`'`).

Steps:

- Selected `Job Skills` column.
- Replaced `[` and `]` with empty string.
- Replaced `'` with empty string.
- Trimmed whitespace using **Transform > Trim**.

Split Job Skills Column into Rows

Split the `Job Skills` column by comma into separate rows for skill-level analysis.

Steps:

- Selected `Job Skills` column.
- Clicked **Split Column** > **By Delimiter** > comma.
- Selected **Split into Rows** under Advanced Options.
- Renamed column to `Skill`.

Screenshots:

	Job Posting ID	Job Posting Date	Job Title	Job Title Full	Job Title Additional Info
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid
	Error 0%	Error 0%	Error 0%	Error 0%	Error
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty
	1000 distinct, 1000 unique	658 distinct, 404 unique	44 distinct, 9 unique	399 distinct, 301 unique	150 distinct, 125 unique
1	2681258907	1/2/2017	Business Intelligence Analyst	Jr. Business Intelligence Analyst	-
2	2723217037	1/4/2017	Developer	Liferay Developer	-
3	2682293940	1/6/2017	Software Engineer	Software Engineer -Remote	-
4	2677255610	1/7/2017	Data Analyst	Senior Business Data Analyst	-
5	2676397282	1/8/2017	Software Engineer	Staff Software Engineer, Backend	-
6	2734084443	1/8/2017	Marketing Analyst	Marketing Analyst	-
7	2671714552	1/9/2017	Software Engineer	Software Engineer	-
8	2712137853	1/18/2017	Data Engineer	Data Engineer	-
9	2680941739	1/20/2017	Product Manager	Product Manager	-
10	2681786037	1/22/2017	Software Engineer	Software Engineer	Customer Experience Services
11	2685275354	1/23/2017	Data Scientist	Senior Data Scientist	-
12	2748686994	1/25/2017	Data Analyst	Data Analyst	-
13	2681782167	1/25/2017	Software Engineer	Software Engineer	-
14	2670486658	1/26/2017	Developer	Salesforce Developer	Cross Train from Java, JavaScript,
15	2682234270	1/28/2017	Data Engineer	Data Engineer	-
16	2680026121	1/29/2017	Data Engineer	Lead Data Engineer	-
17	2713380624	2/3/2017	Data Scientist	Principal Data Scientist	-
18	2714579708	2/5/2017	Software Developer	Full Stack Software Developer	Video Annotation ML Startup
19	2690951888	2/6/2017	Operations Analyst	Sap Business Operations Analyst	-
20	2708347585	2/6/2017	Software Developer	Software Developer	-
21	2723135314	2/8/2017	Senior Analyst	Senior Analyst, Healthcare	-

no profiling based on top 1000 rows

	Job Position Type	Job Position Level	Years of Experience	Job Location	Minimum Pay	Maximum Pay
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
	4 distinct, 1 unique	6 distinct, 0 unique	14 distinct, 1 unique	216 distinct, 103 unique	175 distinct, 41 unique	207 distinct, 43 unique
1	Contract	Mid-Senior level	4	Raleigh, NC	71000	80000
2	Full-time	Mid-Senior level	6	Atlanta, GA	162000	149000
3	Full-time	Entry level	0	Greenwood Village, CO	58000	88000
4	Full-time	Mid-Senior level	6	United States	129000	126000
5	Full-time	Mid-Senior level	4	Palo Alto, CA	100000	150000
6	Full-time	Associate	3	Baltimore, MD	49000	79000
7	Full-time	Mid-Senior level	8	Ellicott City, MD	88000	114000
8	Full-time	Mid-Senior level	3	Charlotte, NC	81000	143000
9	Full-time	Mid-Senior level	8	New York City Metropolitan Area	132000	146000
10	Full-time	Mid-Senior level	4	United States	57000	106000
11	Full-time	Mid-Senior level	5	New York, United States	136000	153000
12	Contract	Associate	4	Dallas-Fort Worth Metroplex	68000	71000
13	Full-time	Associate	3	United States	71000	74000
14	Full-time	Entry level	0	New York, NY	50000	49000
15	Contract	Mid-Senior level	8	Creve Coeur, MO	123000	147000
16	Full-time	Mid-Senior level	7	United States	111000	151000
17	Full-time	Executive	5	United States	190000	177000
18	Full-time	Mid-Senior level	4	Pittsburgh, PA	72000	146000
19	Contract	Mid-Senior level	5	United States	63000	84000
20	Full-time	Entry level	2	San Francisco Bay Area	73000	84000
21	Full-time	Associate	6	San Francisco, CA	66000	113000

WUZZUF

	Pay Rate	Number of Applicants	Company Name	Company Industry	Company Size	Year
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 10 Error Empty
	2 distinct, 0 unique	109 distinct, 36 unique	483 distinct, 337 unique	51 distinct, 17 unique	8 distinct, 0 unique	4 distinct, 0 unique
1	hr	100	OnDemand Agility Solutions	Information Technology & Services	1,001-5,000 employees	.
2	hr	3	ASAP Solutions Group, LLC	Management Consulting	501-1,000 employees	.
3	yr	0	MiTek	Construction	5,001-10,000 employees	.
4	yr	33	Insight Global	Staffing & Recruiting	1,001-5,000 employees	.
5	yr	1	Affirm	Financial Services	1,001-5,000 employees	.
6	yr	0	Brave Thinking Institute	Professional Training & Coaching	11-50 employees	.
7	yr	7	Huntress	Information Technology & Services	51-200 employees	.
8	yr	12	Red Ventures	Internet	1,001-5,000 employees	.
9	yr	72	Prestige Staffing	Staffing & Recruiting	51-200 employees	.
10	yr	10	O'Reilly Auto Parts	Retail	10,001+ employees	.
11	yr	7	Harnham	Staffing & Recruiting	51-200 employees	.
12	hr	46	Apex Systems	Information Technology & Services	1,001-5,000 employees	.
13	yr	0	Insight Global	Staffing & Recruiting	1,001-5,000 employees	.
14	yr	51	Revolent Group	Information Technology & Services	51-200 employees	.
15	hr	7	Revolution Technologies	Staffing & Recruiting	1,001-5,000 employees	.
16	yr	4	Austin Fraser	Staffing & Recruiting	201-500 employees	.
17	yr	45	Harnham	Staffing & Recruiting	51-200 employees	.
18	yr	12	TEEMA	Staffing & Recruiting	201-500 employees	.
19	hr	43	Insight Global	Staffing & Recruiting	1,001-5,000 employees	.
20	yr	180	TEKtalent Inc	Human Resources	201-500 employees	.
21	yr	2	CB Insights	Internet	201-500 employees	

Year	Month	year_month
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
4 distinct, 0 unique	12 distinct, 0 unique	
2017	1	2017-01
2017	1	2017-01
2017	1	2017-01
2017	1	2017-01
2017	1	2017-01
2017	1	2017-01
2017	1	2017-01
2017	1	2017-01
2017	1	2017-01
2017	1	2017-01
2017	1	2017-01
2017	1	2017-01
2017	1	2017-01
2017	1	2017-01
2017	1	2017-01
2017	1	2017-01
2017	1	2017-01
2017	2	2017-02
2017	2	2017-02
2017	2	2017-02
2017	2	2017-02
2017	2	2017-02

Query Settings

PROPERTIES

Name

Wuzzuf-Jobs-Posting

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Replaced Value
- Replaced Value1
- Filtered Rows
- Replaced Value2
- Filtered Rows1
- Filtered Rows2
- Filtered Rows3
- Replaced Value3
- Removed Columns
- Inserted Year
- Inserted Month
- Added Custom
- Removed Columns1

Trim and Clean Skills

Ensure `Skill` column has no extra spaces or residual characters after splitting.

Steps:

- Selected `Skill` column.
- Used **Transform > Trim** and optionally **Clean**.
- Verified skill values are clean.

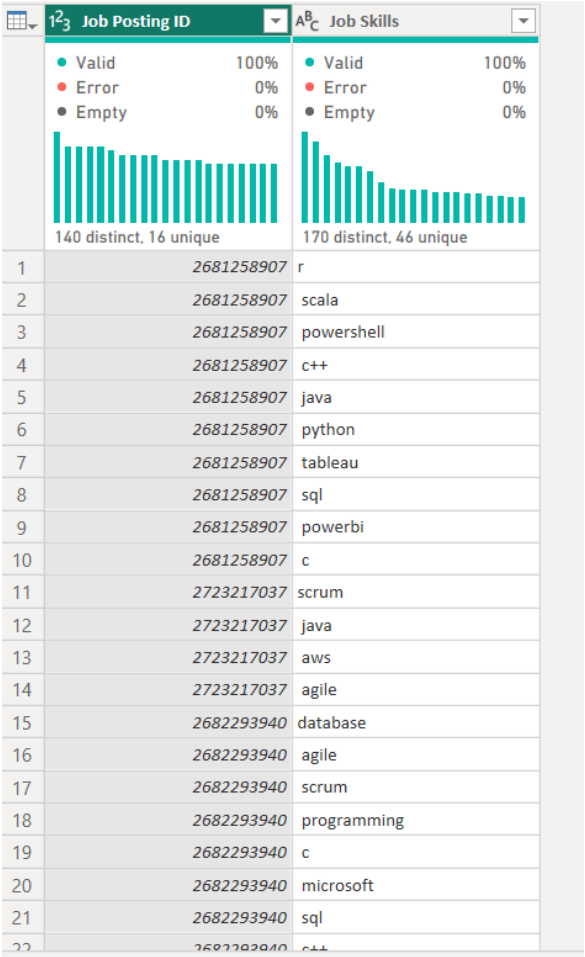
Create Skills Table

Create a dedicated `Skills` table with only `Job Posting ID` and `Skill`.

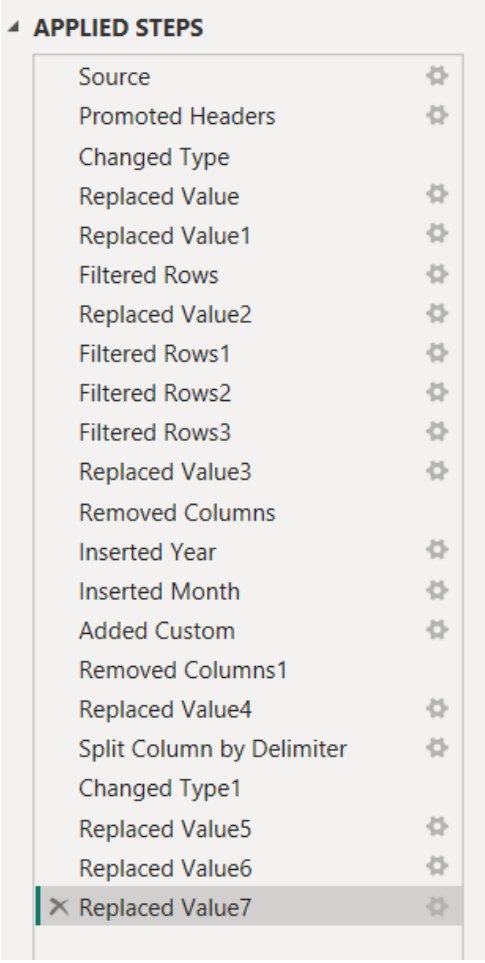
Steps:

- Right-clicked query > **Duplicate**.
- Renamed query to `Skills`.
- Removed all columns except `Job Posting ID` and `Skill`.
- Removed duplicates from `Skills` table.

Screenshot:



	Job Posting ID	Job Skills
	Valid 100%	Valid 100%
	Error 0%	Error 0%
	Empty 0%	Empty 0%
	140 distinct, 16 unique	170 distinct, 46 unique
1	2681258907	r
2	2681258907	scala
3	2681258907	powershell
4	2681258907	c++
5	2681258907	java
6	2681258907	python
7	2681258907	tableau
8	2681258907	sql
9	2681258907	powerbi
10	2681258907	c
11	2723217037	scrum
12	2723217037	java
13	2723217037	aws
14	2723217037	agile
15	2682293940	database
16	2682293940	agile
17	2682293940	scrum
18	2682293940	programming
19	2682293940	c
20	2682293940	microsoft
21	2682293940	sql
22	2682293940	c++



Extract Year and Month from Job Posting Date

Add `Year`, `Month`, and `Year-Month` columns for time-based filtering.

Steps:

- Selected `Job Posting Date`.
- Added `Year` and `Month` columns from **Add Column > Date**.
- Created `Year-Month` using a custom column: `Text.From([Year]) & "-" & Text.PadStart(Text.From([Month]), 2, "0")`.

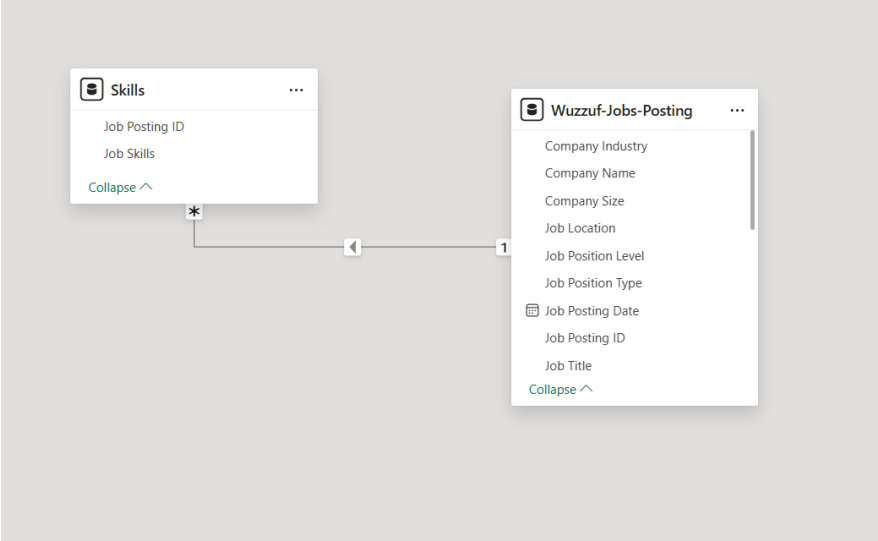
Load Queries to Data Model

Load both `Wuzzuf-Jobs-Posting` and `Skills` queries to Power BI data model.

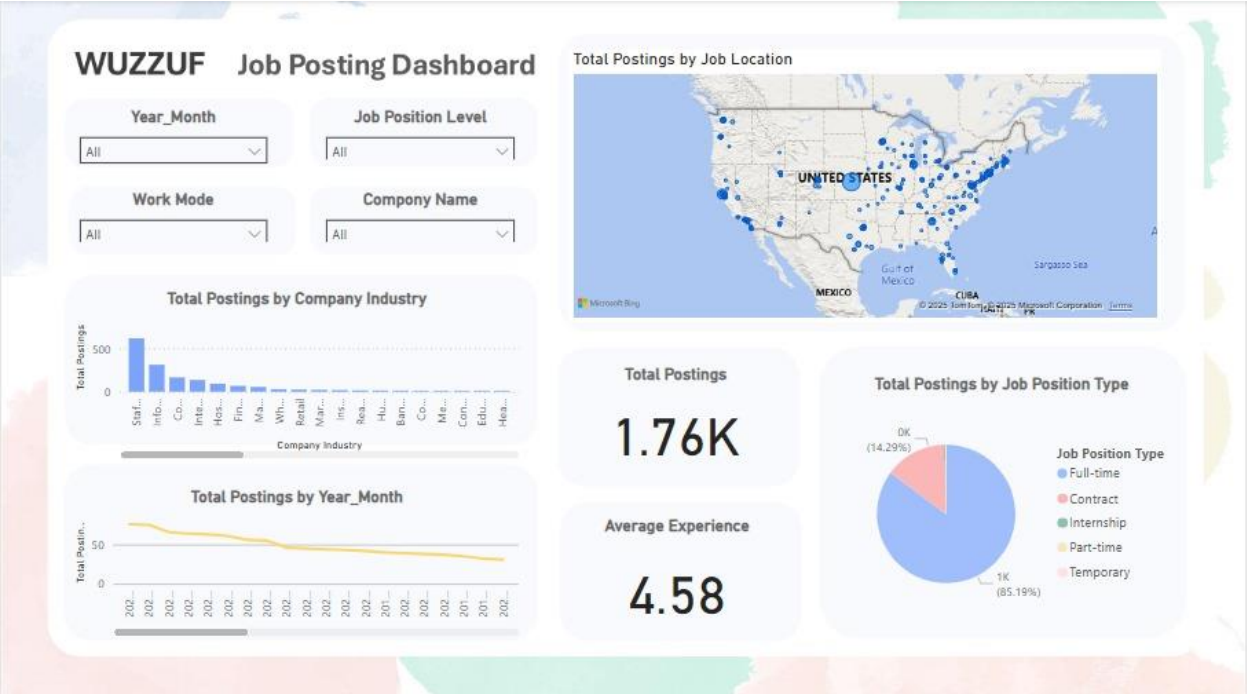
Steps:

- Clicked **Close & Apply** in Power Query Editor.
- Confirmed both queries are added to the model.

Screenshot:



Dashboard:



Notes

- **Dataset**: The `Wuzzuf-Jobs-Posting.csv` file contains 25,116 rows with columns like `Job Posting ID`, `Job Skills`, and `Years of Experience`. Transformations ensure clean, usable data.

- **Performance**: Splitting `Job Skills` into rows increases the `Skills` table size. Monitor query performance and consider filtering by `Job Posting Date` (e.g., 2020– 2021) if slow.

- **Screenshots:** Capture each screenshot in Power BI Power Query Editor, ensuring the **Applied Steps** pane is visible to show the context. Save screenshots as PNG or JPG and insert them here.

- **Deliverable**: Include this documentation and screenshots in the `JobPulse_Deliverables` folder, alongside `Wuzzuf-Job-Posting-Cleaned.xlsx` and `JobPulse-Dashboard.pbix`

Project Link:

https://app.powerbi.com/view?r=eyJrIjoiMDUxZmI0ZmItN2JiMi00MGEyLThlOTQtZDU4N2EyMmFiNDA3IiwidCI6IjhjMWFlYThiLTJmMzQtNDk5NS1hMTY0LTQzYmM3YjZlYTczNyJ9